UK Maths Trust

Pink Kangaroo

Thursday 19 March 2026

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England & Wales: Year 11 or below | *Scotland: S4 or below* | *Northern Ireland: Year 12 or below*

Instructions

1. Do not open the paper until the invigilator tells you to do so.
2. Time allowed: **60 minutes**.
   No answers, or personal details, may be entered after the allowed time is over.
3. The use of blank or lined paper for rough working is allowed; **squared paper**, **calculators and measuring instruments are forbidden**.
4. **Use a B or an HB non-propelling pencil.** Mark at most one of the options A, B, C, D, E on the Answer Sheet for each question. Do not mark more than one option.
5. **Do not expect to finish the whole paper in the time allowed.** The questions in this paper have been arranged in approximate order of difficulty with the harder questions towards the end. You are not expected to complete all the questions during the time. You should bear this in mind when deciding which questions to tackle.
6. **Scoring rules:**
   5 marks are awarded for each correct answer to Questions 1-15;
   6 marks are awarded for each correct answer to Questions 16-25;
   In this paper you will not lose marks for getting answers wrong.
7. **Your Answer Sheet will be read by a machine.** Do not write or doodle on the sheet except to mark your chosen options. The machine will read all black pencil markings even if they are in the wrong places. If you mark the sheet in the wrong place, or leave bits of eraser stuck to the page, the machine will interpret the mark in its own way, or reject the answer sheet.
8. **The questions on this paper are designed to challenge you to think, not to guess.** You will gain more marks, and more satisfaction, by doing one question carefully than by guessing lots of answers. This paper is about solving interesting problems, not about lucky guessing.

Enquiries about the Pink Kangaroo should be sent to:

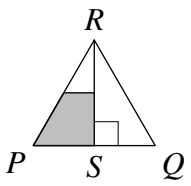
challenges@ukmt.org.uk *www.ukmt.org.uk*

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1. Which of these expressions has the smallest value?

A $202 \div 6$ B 202.6 C $20 + 26$ D $202 - 6$ E 20×26

2. Amber draws an equilateral triangle PQR. The point S is the midpoint of PQ. A line is drawn parallel to the base, PQ, through the midpoint of RS.



What fraction of the area of triangle PQR is shaded?

A $\frac{1}{8}$ B $\frac{3}{10}$ C $\frac{1}{4}$ D $\frac{3}{8}$ E $\frac{1}{3}$

3. Yong writes down a 7-digit number, '193391a'. His number is divisible by 6. What is the value of a?

A 0 B 2 C 4 D 6 E 8

4. What is the value of the expression $(1 - 2) - (3 - 4) - (5 - 6) - \ldots - (2025 - 2026)$?

A -1013 B -1011 C 1011 D 1013 E 2024

5. Sanika wants to write the numbers 1, 2, 3, 4, 5, 6 and 7 in the boxes in the diagram. She has already written two numbers, as shown.



She wants the sum of the numbers in each pair of adjacent boxes to be odd, and to avoid the sum of the numbers in any three consecutive boxes being a multiple of 3.

What is the sum of the numbers she will need to write in the shaded boxes?

A 5 B 7 C 9 D 11 E 13

6. A farmer has dogs, sheep, goats, pigs and chickens on her farm. There are more chickens than pigs, more pigs than goats, more goats than sheep and more sheep than dogs. There are half as many dogs as chickens. The total number of animals is as small as it could be. How many animals are there on her farm?

A 28 B 30 C 32 D 34 E 36

7. After a day trip through the Scottish Highlands, five hikers are covered in midge bites.

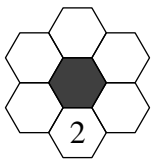
They have 7, 9, 10, 13 and 14 bites. The total number of bites Gabriel and Jesseca have is three times the number of bites Isobel has.

The total number of bites Kyle and Jesseca have is twice the number of bites Henry has.

How many bites does Jesseca have?

A 7 B 9 C 10 D 13 E 14

8. The six smallest prime numbers are to be written in the empty hexagons on the right. Numbers in adjacent empty hexagons must **not** add to a prime number. The 2 is already in place. In how many ways can the hexagons be filled in?

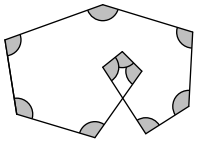


A 2 B 6 C 12 D 60 E 120

9. Fifteen points are equally spaced around a circle. How many regular polygons can be drawn by selecting the vertices of each polygon from these points?

A 5 B 7 C 9 D 11 E 13

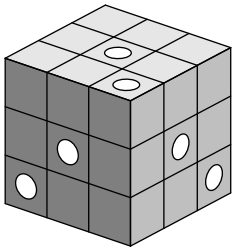
10. The diagram, which is not drawn to scale, shows a shape with ten equal angles marked. What is the size of each of these angles?



A 96° B 105° C 108° D 115° E 120°

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11. Six woodworms made their home in an old wooden cube made up of identical small cubes. Each one drilled a tunnel all the way through the cube, parallel to one of its edges. The diagram shows the entrances to the six tunnels.



How many small cubes do **not** have a tunnel drilled through them?

A 9 B 12 C 15 D 18 E 21

12. Five pupils took part in a one-mile run. One of them didn't finish and the others finished at different times. When asked about the race later, they said:

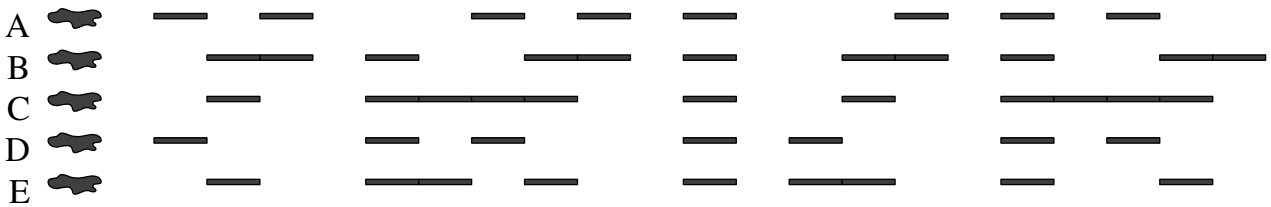
Priya: "I was second or third."
Qusay: "I didn't make it to the finish line."
Rathursan: "I got to the finish line and wasn't fourth."
Sara: "I was fourth."
Tanisha: "I was first."
One of the pupils lied and all the others told the truth. Which of the pupils lied?

A Priya B Qusay C Rathursan D Sara E Tanisha

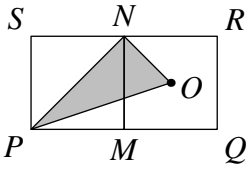
13. Donita rides her penny-farthing through a small puddle, as shown in the picture below.



What might the track that she leaves behind look like?

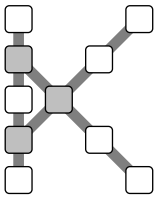


14. Let M and N be the midpoints of PQ and RS respectively in the rectangle $PQRS$. Let O be the centre of the rectangle $MQRN$. What fraction of the rectangle $PQRS$ is shaded?



A $\frac{1}{4}$ B $\frac{1}{2}$ C $\frac{1}{3}$ D $\frac{1}{5}$ E $\frac{2}{9}$

15. Avani wants to place the numbers 1 to 10 in the boxes of the K-shaped grid shown. She wants the sum of the numbers in each line of boxes - either 5 in a row vertically or 4 in a row diagonally - to be the same.
She also wants this sum to be as large as possible.
What will be the sum of the numbers she places in the three shaded boxes.



A 13 B 18 C 23 D 26 E 27

16. In the first round of a chess tournament, each player plays against every other player exactly once. A player gets 3, 1 or -1 points respectively for winning, drawing or losing a match. At the end of the first round, the sum of the scores of all the players is 90. How many players are there in the tournament?

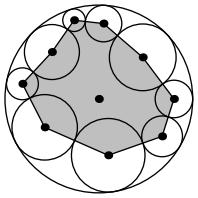
A 10 B 11 C 12 D 13 E 14

17. Given that, $2026 = a^b - ab$ for two positive integers a and b. What is the value of $a + b$?

A 10 B 13 C 15 D 1013 E 1015

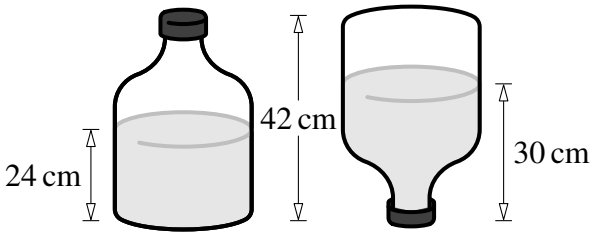
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18. The diagram shows a large circle of radius 10 and nine smaller circles, each tangential to two other small circles and also tangential to the large circle. The sum of the distances between the centres of the smaller circles and the centre of the large circle is equal to d.
What is the perimeter of the shaded polygon in terms of d?



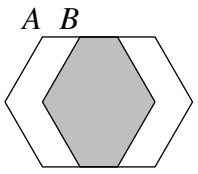
A $90 - 2d$ B $90 - d$ C $180 - d$ D $180 - 2d$ E $180 + 2d$

19. Michael has a bottle partially filled with water. The figure on the right shows how the height of water in the bottle changes when it is flipped upside down. The bottle's capacity is 4.5 ℓ and the water-filled part in the first diagram is cylindrical.
What is the volume of water in the bottle?



A 2.1 ℓ B 2.4 ℓ C 2.5 ℓ D 2.7 ℓ E 3.0 ℓ

20. Two regular hexagons with side-lengths of 60 are shown in the diagram which is not drawn to scale. The right-hand hexagon was created by shifting the left-hand hexagon horizontally by the length of the segment AB. This creates three regions of equal area. What is the length of AB?

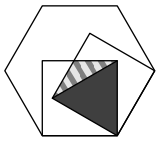


A 30 B 39 C 40 D 45 E 52

21. Nico has eight sticks with distinct integer lengths, no three of which are able to form a triangle. What is the shortest possible length of the longest stick?

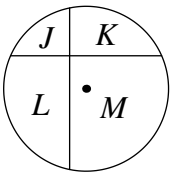
A 30 B 31 C 32 D 33 E 34

22. Aadi draws a regular hexagon and two squares inside the hexagon, as shown. What is the ratio of the area of the striped triangle to the area of the shaded triangle?



A 1 : 4 B $\sqrt{3} : 6$ C 1 : 3 D 2 : 5 E 1 : 2

23. Two perpendicular chords are drawn in a circle with radius 12 cm, dividing the circle into four regions, as labelled in the diagram. One chord lies 3 cm from the centre, and the other lies 4 cm from the centre. The sum of the areas of regions J and M is X cm^2 larger than the sum of the areas of regions K and L. What is the value of X?

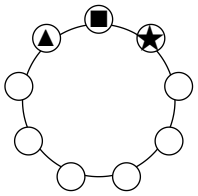


A 9 B 16 C 36 D 48 E 60

24. Elijah and Mason alternate in taking toffees out of a box: Elijah takes 1, then Mason takes 2, then Elijah takes 3, then Mason takes 4, and so on. When there are not enough toffees to follow this pattern, the boy whose turn it is takes all of the remaining ones. At the end, Elijah has 407 toffees. How many toffees were in the box at the beginning?

A 814 B 827 C 834 D 841 E 851

25. Anni places the digits $1, 2, \ldots, 9$ on a circle in some order. She reads three adjacent digits clockwise to form a three-digit number, such as ▲ ■ ★ as in the diagram, and writes down all nine such numbers. One of these numbers is a, which is a factor of the sum of the remaining 8 numbers. How many possible values of a are there?



A 1 B 2 C 3 D 4 E 5

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